CHAPTER 21

Impact of microplastics on economic condition in underdeveloped nations

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21.1 Introduction

Microplastic are small plastic particles that are produced by the disintegration of plastic particles. The use of plastics and its products has risen in the past few decades. They are used across almost all fields including electronics, the paper industry, the manufacture of toys, agricultural and household products, cosmetics, the packaging field, the food sector, etc. Currently, the manufacture of plastics and its products has risen by 6% annually. As per a recent study, global plastic pollution has increased up-to 16%, i.e., approximately 60 million tons of plastic yearly are produced (Gasperi et al., 2018). Once the plastic is used and discarded, it reaches the nearby water bodies and breakdown into smaller particles known as macro plastics, meso-plastics, microplastics, and nano-plastics based on their sizes. The size of these microplastics ranges from less than 5 mm (Karthikeyan & Subagunasekar, 2022; Windsor et al., 2019). The impact of these microplastics is an emerging threat to both humans as well as marine species. These plastics are ubiquitous and found across the oceans, seashores, lakes, rivers, and land areas as well (Horton, Walton, et al., 2017).

21.2 Sources and occurrence of microplastics

Microplastics are formed by the fragmentation of large plastic particles. The physical, chemical, or biological process acted on the primary and secondary plastics form the microplastics ranging from 1 μm to 5 mm in size (Cole et al., 2011; Dawson et al., 2018; Frias & Nash, 2018; Horton, Walton, et al., 2017; Karthikeyan & Subagunasekar, 2022; Rodríguez-Hernández et al., 2020). The first plastic pollution was recorded in the 1970s from then the contamination increased day by day as the applications of plastics are rising. The arise of

Microplastics.
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plastic pollution in seas and oceans is mainly from the terrestrial environment (Andrady, 2011). Sometimes freshwater can also be one of the main transporters of plastic debris from the terrestrial environment to the seas and oceans (Iannilli et al., 2023). A few of the significant sources of microplastics can be agricultural or road runoff, untreated stormwater, and tributaries into the nearby lakes or rivers. Rain and storms can be a key factors causing the movement of these plastics into the water eco-system (Ballent et al., 2016; Hitchcock, 2020). The release of waste from factories, or the cargo losses caused during shipping can cause the release of plastics into the surrounding water bodies (Sri & Mishra, 2023) . Many reports also show that the direct release of plastics into the water environment can be via commercial fishing or even through wind and tidal activities (Hoffmann & Reicherter, 2014; Lusher et al., 2015; Browne et al., 2007; Horton, Svendsen, et al., 2017).

Microplastics are transported and they reach the drinking water system as well. A study by Maocai Shen et al. (2021), showed that more than 70% of the water samples contained polymers, which were composed of polyethylene, polypropylene, and polyethylene terephthalate (Fig. 21.1).

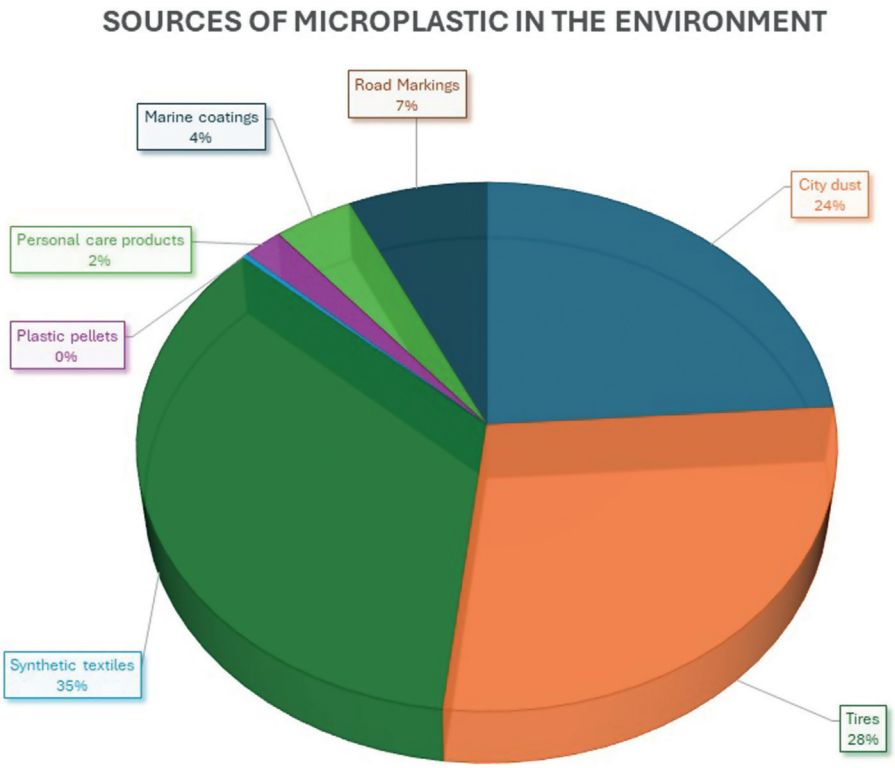

Figure 21.1
Sources of microplastic in the environment.

21.3 Mismanagement of plastic waste

As microplastics are rising every day, managing waste and its pollution is an important process required currently. A study by Lebreton and Andrady (2019), stated the mismanagement of plastic waste can lead to more disasters which by 2060 can raise up to 155–265 million t. A study by Abelouah et al., showed that microplastics found across the Atlantic coast had different forms and shapes among which fragments, fibers, films, and pellets were common microplastics found (Abelouah et al., 2022). Also, this study discusses various percentages of all kinds of plastics that were found across the shoreline of Morocco among which polyethylene followed by polypropylene were the highest microplastic present. Waste plastics in the ocean's environment range from approximately 4.8 to 12.7 Mt/year, i.e., 1.4%–3.6% of current global production. A recent study also shows that synthetic plastic produced is 6.3 billion metric per ton also 1.4%–3.6% of annual production of plastic has been lost to the oceans (Geyer et al., 2017). Globally, this is one of the significant problems causing economic loss in many countries as well. There is a strong correlation between social factors and a country's level of economic development and industrialization, and these factors may have an indirect influence on the amount of microplastics present (Wang, 2018). According to the UN Environmental Programme, developed countries are generally less polluted by microplastics than developing countries. Table 21.1 shows that developed countries have a small population, a high level of urbanization, and a lower contribution rate to secondary industry than developing countries.

21.4 Status and prevalence of microplastics pollution in underdeveloped countries

Plastic has become an indispensable part of modern life, but its environmental impact, particularly its non-biodegradability, has led to the widespread presence of microplastics, which are tiny plastic particles found in various environments (Menon et al., 2023). The presence of microplastics has been documented in diverse locations, including water bodies,

Table 21.1: Comparison of the economic development and industrial structure of developing countries and under-developed countries.

	Developed country	Under-developed countries
Number of waterbodies	20	17
Number of countries involved	5	7
Average population (10,000 people)	33,279.66	8552.86
Average population density (n/km^2)	47.82	156.42
GDP per capita (10,000 yuan)	60.10	356.72
Average secondary industry contribution rate (%)	22.67	19.67
Average urbanization rate (%)	63.90	80.72

soil, and even the atmosphere (Koutnik et al., 2021). This has raised concerns about the potential impact on human health and the environment, especially in countries with high plastic consumption and inadequate waste management systems (Ayeleru et al., 2020).

In India, the illegal dumping of plastic waste into waterways is a common practice, leading to a substantial amount of plastic ending up in rivers and lakes (Bhargava, 2022). Studies have reported high concentrations of plastic debris in Indian coastal areas, emphasizing the growing problem of microplastic contamination in the country (Amrutha et al., 2021). Despite being a major producer of plastic waste, India's research on microplastics is still limited compared to other regions, highlighting the need for more comprehensive studies to understand the extent of microplastic pollution in the country (Vaid et al., 2021).

In Bangladesh, the inadequate enforcement of plastic waste management laws has resulted in a significant increase in plastic waste, with a substantial portion ending up in landfills. The use of polythene bags, despite being banned, remains prevalent due to the lack of affordable and environmentally friendly alternatives (Hossain et al., 2021). In Pakistan, the coastal areas and the Indus River have been heavily impacted by the dumping of large amounts of plastic waste, with a significant proportion of the country's total waste being composed of plastic. The government has taken steps to address this issue by banning the use of plastic bags in several regions (Irfan et al., 2020). In China, the widespread use of plastic in various industries, such as the express delivery and agriculture sectors, has contributed to microplastic pollution. The annual release of microbeads into the environment is a significant concern, highlighting the need for better management of plastic waste in the country (Fu & Wang, 2019). Brazilian plastic production increased by 57% in 2018, with 8.3 million tons being produced in 2018 and Brazil recycles only about 1.3% of its plastic waste, indicating a potential for contamination of the environment by these residues (Bertoldi et al., 2021; World Wildlife Fund, 2019). It may be reasonable to assume that many sources release plastics into Lake Guab, Brazil, judging by the high abundance of colored microplastics. Additionally, white-transparent microplastics might be responsible for the high frequency of this type of microplastic due to the fading of colored plastics caused by UV light and other environmental stressors (Bertoldi et al., 2021; Drabinski et al., 2023).

Between 1990 and 2007, an examination of records from 33 out of 54 African nations reveals a notable plastic consumption rate of approximately 172 million tons of polymers and plastics, valued at $285 billion (Babayemi et al., 2019; Okeke et al., 2022). Despite this substantial consumption, inadequate waste management policies persist, causing Africa to lead globally in terms of mismanaged plastic. Notably, Africa possesses the smallest percentage of freshwater bodies (9%) among continents, emphasizing the need for sustainable management (Alimi et al., 2021). The burgeoning issue of microplastic pollution further exacerbates challenges faced by the African population. Although the initial study

reported microplastics (undefined in size at that time) originated in Africa, microplastics research on the continent remains in its nascent stages (Balcom, 2022). While awareness of microplastic pollution has heightened in the last decade, the occurrence, distribution, and fate of microplastics in diverse environmental compartments within Africa remain poorly understood (Shahul Hamid et al., 2018).

Drawing from current literature, microplastics distribution in surface waters, sourced from both marine and freshwater environments, has been documented in only 5 out of 54 African countries, constituting approximately 9.3% coverage (River, 2022). In Nigeria, for instance, sampling during the rainy season revealed higher microplastic abundance compared to the dry season, echoing a pattern observed in River Nakdong, South Korea, where 15,560 particles/m^3 were reported during the rainy season versus 1410 particles/m^3 in the dry season (Schnurr et al., 2018; Tsang et al., 2017). A parallel study in Orange-vaal River, South Africa, conducted by Weideman et al. (2019), similarly affirmed an increase in microplastic abundance during the rainy season (Weideman et al., 2019). Nadioo and Glassom (2019) reported a twofold rise in microplastic abundance during the winter season compared to summer in marine waters along the coastlines of South Africa (Naidoo & Glassom, 2019). Conversely, a report from Kenya indicated the lowest microplastic abundance in the surface waters of Naivasha lake, with an average value of 0.407 ± 0.135 particles/m^3, predominantly composed of polypropylene, polyethylene, and polyester (Alimi et al., 2021).

This investigation reveals an exceptionally high prevalence of microplastics in sediments, representing one of the most substantial documented occurrences globally. The elevated concentrations are attributed to intense marine and industrial activities in the vicinity, particularly notable in a region ranked second in seafood product exports (Abidli et al., 2017). Sampling sites manifesting heightened microplastic levels include the Beibu Gulf and the coastline of the China Sea, where concentrations reached 8714 particles/kg (Qiu et al., 2015). Similarly, the Canadian Lake Ontario nearshore exhibited concentrations ranging from 20 to 27,830 particles/kg (Ballent et al., 2016), while the Kachelotplate registered an alarming 62,100 particles/kg (Liebezeit & Dubaish, 2012). Considering these findings, the omnipresence of microplastics presents a substantial threat to both the environment and human health. Urgent and decisive measures are imperative to address the escalating predicament of plastic pollution and mitigate its consequential impacts.

21.5 Economic impacts of microplastics pollution

21.5.1 Impact on agriculture

Rural areas, especially agricultural soils, face a significant threat from microplastic pollution, posing potential risks to food security, biodiversity, and the overall well-being of rural communities (Mihai et al., 2021). Agroecosystems, vital for sustaining food

production, have already been stressed by climate change and population growth. According to Zhou et al. (2020), the existence of microplastics in soils creates an extra layer of stress that requires in-depth knowledge to evaluate its effects on rural people, the global food supply, and the environment. Because they depend so largely on the land, rural populations may experience negative environmental and economic effects from pollutants such as microplastics (Usman et al., 2022). The Food and Agriculture Organisation (FAO) stresses the value of having healthy soils and the financial consequences of lower crop yields and quality because of pollution. Though studies on terrestrial microplastics provide important information, there is a dearth of research on the direct effects of plastic contamination on rural people (Joint, 2022).

Current knowledge identifies diverse sources of microplastic pollution in agricultural soils, influenced by land use and farming practices. These sources include the application of microplastics through farming activities (e.g., sewage sludge, compost, and wastewater irrigation), degradation of macroplastics used in farming (e.g., mulching films) (Fakour et al., 2021; Zhang et al., 2020), indirect application through activities like openly burning plastic garbage, doing laundry, and moving stuff from other environmental sections (e.g., freshwater bodies and road-related particles) (Rajmohan et al., 2019; Wang et al., 2022).

Farming practices, such as sewage sludge application, contribute significantly to microplastic abundance in agroecosystems. Plastic films and sheeting in agriculture, while enhancing crop production, can become sources of persistent microplastic pollution if not properly managed (Corradini et al., 2019; Okeke et al., 2022). The long-term retention of microplastics in soil can lead to soil degradation and water repellence, impacting crop yield and quality, albeit quantifying this impact remains a challenge (Yang et al., 2020). Microplastic pollution in rural soils exhibits dynamic movement, with potential transfers between soil and water. Soil erosion and various transport mechanisms, including aeolian transport and vertical distribution, contribute to the complexity of microplastic movement within the environment. Agricultural practices play an important role in the spread of microplastics in rural lands, necessitating further investigation (Kallenbach et al., 2022; Zeng, 2023).

While studies indicate widespread microplastic pollution in rural soils globally, the actual concentrations and geospatial distribution remain insufficiently understood. Recent research has focused on the impact of microplastics on the flora and fauna of terrestrial ecosystems, revealing adverse effects on detritus feeders, such as earthworms, as well as negative impacts on various plants. These findings underscore the potential threats to rural livelihoods and crop production (Qaiser et al., 2023; Wang et al., 2022). To advance our understanding, further research is needed in quantifying and identifying the sources of microplastics through extensive soil sampling in agricultural and rural areas. Current variations in microplastic concentrations used in studies highlight the necessity for

standardized and environmentally relevant amounts. To identify possible risks to soil, water, the agri-food industry, and human health at local and regional levels, a thorough impact analysis of terrestrial microplastics in rural areas is essential (Mihai et al., 2021).

21.5.2 Impact on water resources and aquatic ecosystems

Divergent observations characterize the presence of various plastic types in organisms across multiple studies, with a significant prevalence of microfibers consistently reported. A recent investigation in South Africa revealed that 87% of seven fish species harbored microfibers, constituting 95% of the total microplastics identified. However, detailed information on the composition of these fibers was not provided (Sparks & Immelman, 2020). Other studies in the same region documented elevated fiber levels, with one reporting 67% in microplastics collected from mussels, and another indicating that 97% of microplastics recovered from mussels in Tunisia were in fiber form (Abidli et al., 2019; Sparks, 2020). Akindele and Alimba (2021) found that fibers predominated among the recovered plastic types from insects in Nigeria. These findings align with Rochman et al.'s (2015) identification of microfibers as a dominant plastic-type in aquatic biota (Akindele & Alimba, 2021; Rochman et al., 2019). Moreover, a variety of polymer types have been identified in organisms under examination, deviating notably from those recovered from surface waters and sediments (Lu et al., 2021). Polyethylene emerges as the predominant plastic-type across all organisms, with polypropylene following suit (Davis, 2022). This hierarchical occurrence mirrors the plastic composition observed in surface water analyses.

21.5.3 Consequences on food supply chains

In 2020, food packaging constituted the largest share (39.6%) of plastic usage in Europe, encompassing a variety of materials such as plastics, metals, paper, board, glass, adhesives, and coatings (Mihai et al.,2021). The proliferation of retail industries in urban areas has extended the reach of plastic-packaged food to surrounding rural communities, leading to a shift in consumption culture (Ncube et al., 2021). This shift, particularly in peri-urban regions and metropolitan areas, raises concerns about the environmental and public health implications, with rural-urban interactions warranting further investigation.

The public's health may be in danger due to solvent and accidental substance residues found in plastic packaging. 906 compounds were linked to plastic packaging by Groh et al. (2019); of these, 63 were classified by the European compounds Agency as dangerous to human health and 68 as detrimental to the environment. Because of chemicals and additives, composite materials—including plastic—used in food-contact products present dangers. Reuse, disposal, and recycling may cause these chemicals to migrate into food, cosmetics, or the environment (Undas et al., 2023). There is a chance that all plastic

polymers include dangerous chemicals and additives that could migrate into food. For example, it has been discovered that food might contain levels of bisphenol-A (BPA), which is used in polyvinyl chloride, above the limits set by the European Union (Muzeza et al., 2023). Coastal rural communities engaging in fishing and aquaculture may face plastic pollution risks, as demonstrated by the migration of phthalate esters from packaging materials into seafood (Karbalaei et al., 2018). The risks associated with plastic food packaging extend beyond chemical migration (Wrona & Nerin, 2019). Micro- and nano-plastics, unintentionally produced during the manufacturing process, can contaminate food. Such particles, present in marine, freshwater, and terrestrial ecosystems, may adversely affect the health, feeding, growth, and survival of organisms across trophic levels (Benson et al., 2022; Rathod et al., 2023).

In low-income developing countries, the rise in individually packaged plastic-covered fast-consumption foods exacerbates plastic waste issues. Insufficient waste management infrastructure leads to environmental pollution, increasing health risks. In some former Soviet countries, the decline of glass recycling facilities has contributed to the agri-food sector's reliance on packaging with plastics, leading to heightened plastic contamination in rural areas (Mihai et al., 2021). Food resources in rural areas, which frequently practice subsistence farming, may get contaminated by microplastics from both marine and terrestrial ecosystems (Bennett et al., 2023). There is still a lot we don't know about how plastic waste affects rural ecosystems and human health. Furthermore, the health concerns associated with plastic pollution in food are little understood in Asia and Africa, where low-income nations prevail (Velis & Cook, 2021). This calls for more research and attention.

21.5.4 Influence on tourism and recreational activities

Rural tourism encompasses diverse activities in rural areas, including ecotourism, agritourism, and cultural tourism. Economically, it serves as a crucial revenue source, aiding underdeveloped rural infrastructure (Ćurčić et al., 2021). Environmentally and socially, it supports conservation and cultural understanding. Aligned with the UN's Sustainable Development Goals (SDGs), rural tourism contributes to poverty reduction, gender equality, economic growth, innovation, reduced inequalities, sustainable communities, responsible consumption, and environmental protection (Spenceley & Rylance, 2019). In rural areas, inadequate waste management, compounded by a lack of recycling infrastructure, presents a significant challenge. Limited data on solid waste exacerbates the issue, hindering effective waste management strategies. While rural life traditionally generated less waste, increased purchasing power, and packaged goods availability contribute to pollution and water contamination. Tourists, unaware of rural waste management issues, unintentionally exacerbate environmental degradation.

The presence of waste compromises the pristine environment anticipated by tourists, impacting their experience, and discouraging revisits (Ncube et al., 2021; Phelan et al., 2020).

Tourism planning lapses lead to improper waste disposal by tourists, creating an additional environmental burden. Diminished tourist interest in polluted areas affects the tourism industry economically (Aayog, 2018). Well-developed tourist destinations often exhibit stark contrasts between well-maintained tourist spots and polluted local areas (Hooper & Lennon, 2016). This division requires investments in spatial planning to address intricate challenges and prevent environmental pollution. Increased tourist numbers necessitate meticulous planning and robust waste management services to avoid exposing visitors to pollution. Without such measures, the expansion of tourist accommodations can lead to unintended environmental consequences. Rural tourism, while economically beneficial, requires careful attention to spatial planning and sustainability (Dijkstra et al., 2021). Strategic planning ensures positive impacts on the community, fostering a symbiotic relationship between tourism and the environment, and promoting economic growth while preserving ecological integrity (Baloch et al., 2023).

21.5.5 Impact on fisheries and coastal economies

The ingestion of marine plastics by organisms poses a significant threat to commercial fisheries and aquaculture, leading to reduced efficiency and productivity due to physical entanglement and damage. This hazard directly endangers fish stocks, impacting the entire food chain both directly and indirectly. Seafood, comprising over 20% of the food intake by weight for 1.4 billion people (19% of the global population), serves as a primary source of animal protein (Unuofin & Igwaran, 2023). Marine plastics also have substantial effects on recreational activities, with litter and debris polluting coastlines, discouraging tourists, and causing socio-economic challenges (Watkins et al., 2015).

The economic development of coastal areas, vital for a thriving tourism industry, is hindered by plastic pollution, diminishing recreational values, and posing threats to social, human, and mental stability (Aretoulaki et al., 2021). Moreover, the economic ramifications of plastic pollution extend beyond tourism, encompassing health risks. The ubiquitous presence of disposable plastics in the environment, characterized by complex chemical compositions, presents unprecedented challenges in waste management, exacerbated by urbanization (Kibria et al., 2023). Waste governance is undergoing global evolution, with formal municipal or regional waste management prevalent in developed countries, where higher waste generation contrasts with developing nations, often reliant on biogenic products and generating less waste (Maalouf & Agamuthu, 2023). Addressing this issue necessitates the integration of the informal waste sector in developing cities, reduction of consumption in developed areas, standardization of solid waste data collection, and implementation of effective waste management strategies to safeguard both people and the environment.

21.6 Strategies used by developed countries to overcome microplastic pollution and problems associated

21.6.1 Case studies of developed countries

In the realm of environment science, the management of plastic pollution has emerged as a critical concern, especially in developed counties where industrialization and urbanization have significantly contributed to the proliferation of plastic waste. Case studies from advanced nations showcase a multifaceted approach to tackling this global issue, emphasizing the importance of sustainable practices and innovative technologies.

One notable example is Sweden, where a holistic waste management system has been implemented to minimize the environmental impact of plastics. The country employs cutting-edge recycling technologies, including advanced sorting techniques and state-of-the-art recycling plants. Sweden's success lies in its commitment to reducing landfill reliance and promoting a circular economy, where plastic materials are recycled and reused, mitigating the need for single-use plastics (Bucknall, 2020).

In tandem, Germany has implemented stringent regulations and policies targeting plastic production and consumption. The country has embraced a proactive stance by encouraging the use of eco-friendly alternatives, promoting research and development in biodegradable plastics, and imposing levies on non-recyclable materials. Germany's approach reflects a harmonious blend of legislative measures and public awareness campaigns aimed at fostering a culture of responsible plastic use.

Despite these advancements, the insidious presence of microplastics remains a global concern. In countries like Japan and Switzerland, extensive research is underway to comprehend the intricate dynamics of microplastic pollution in aquatic ecosystems. Advanced analytical techniques, such as mass spectrometry and high-performance liquid chromatography, are employed to trace and quantify microplastic particles, providing valuable insights into their distribution and potential ecological ramifications.

In conclusion, the management of plastic pollution in developed countries exemplifies a harmonious convergence of regulatory frameworks, technological innovation, and public engagement. The ongoing quest to understand and address microplastic pollution underscores the collective commitment of the global scientific community toward safeguarding the planet's ecosystems.

21.6.2 Overview of microplastic pollution in developed counties

Microplastic pollution poses a significant environmental challenge in developing countries, where industrialization and modern lifestyles contribute to the widespread dissemination of these microscopic synthetic particles. In the United States, recent studies have indicated an

alarming prevalence of microplastics in freshwater ecosystems, with estimates suggesting concentrations ranging from 0.1 to 1.5 particles/L in various water bodies. Similarly, the United Kingdom has witnessed a surge in microplastic contamination, particularly in urban estuaries and coastal regions, with reported concentrations surpassing 5000 particles/m^3. Japan, despite its stringent waste management policies, grapples with microplastic pollution in its marine environments, recording concentrations exceeding 50 particles/kg of sediment in some coastal areas. Germany, renowned for its recycling initiatives, faces challenges as well, with microplastic concentrations exceeding 500 particles/m^2 in certain river sediments.

21.6.3 Successful initiatives and policies

Controlling microplastic pollution is a complex challenge that demands a combination of regulatory measures, technological innovations, and public engagement. Several policies and initiatives are currently underway globally to address this pervasive issue: (Centers for Disease Control and Prevention CDC, n.d.; European Parliament's Committee on the Environment, n.d.)

- *Ban on microbeads*: Many countries have implemented or are in the process of implementing bans on microbeads in personal care products. Microbeads, often found in exfoliating scrubs and toothpaste, can contribute significantly to microplastic pollution. The bans aim to restrict the use of such materials, subsequently reducing their introduction into aquatic ecosystems.
- *Waste management and recycling policies*: Enhanced waste management strategies, including improved recycling infrastructure, are crucial for preventing plastic waste from entering the environment. Countries are adopting measures to increase recycling rates, promote the circular economy, and discourage single-use plastics.
- *Legislation on single-use plastics*: Some nations have introduced legislation to regulate or ban certain single-use items, such as bags, straws, and utensils. These measures aim to decrease the overall production and consumption of plastics, thereby minimizing the potential for microplastic pollution.
- *Research and monitoring programs*: Governments and environmental agencies are investing in research and monitoring programs to assess the extent of microplastic pollution in different ecosystems. These initiatives provide valuable data to shape effective policies and interventions.
- *International collaborations*: Given the transboundary nature of microplastic pollution, international collaboration is essential. Organizations and agreements, such as the United Nations Environmental Programme (UNEP) and regional bodies, facilitate cooperation among nations to develop unified strategies for mitigating microplastic pollution (Joint Research Centre, n.d.).
- *Innovation in waste reduction technologies*: Ongoing research focuses on developing innovative technologies to capture and remove microplastics from water bodies.

These technologies range from advanced filtration systems to the development of biodegradable materials that pose fewer risks to the environment.

- *Education and awareness*: The occurrence of more marine litter on beaches used by people with limited literacy skills in Brazil and the refusal of microbead products by those who saw awareness campaigns highlight the power of education in combating (micro)plastic pollution. However, until recently, there was limited information on microplastic pollution, and 73% of students in Chile were not aware of the problem. Media features like BBC's Blue Planet II and National Geographic's "Plant or Plastic," along with apps such as The Marine Debris Tracker and SeaCleaner, and free massive open online courses like the MOOC on Maine Litter, as well as lectures and activities from platforms like TechWild and The Oceans Nova Scotia, all contribute to the increasing interest in this environment issue. These educational tools play a crucial role in spreading awareness and knowledge about microplastic pollution.

By combining these policies and initiatives, there is hope for mitigating the impacts of microplastic pollution and fostering a more sustainable relationship with plastic materials. Some of the policies made over the years are presented in Table 21.2 (EU, 2021).

21.6.3.1 European Union

The European Union (EU) has been at the forefront of efforts to address microplastic pollution through a series of legislation and regulations. Some notable measures include: (European Chemicals Agency ECHA, n.d.; European Commission's, n.d.; (European Environment Agency EEA, n.d.)).

21.6.3.1.1 Legislation and regulations

- Single-Use Plastics Directive: The European Union (EU) implemented the Single-Use Plastics (SUP) directive targets products such as plastic cutlery, straws, plates, and cotton buds, mandating member states to enact measures for reducing their consumption and ensuring appropriate labeling (EU, 2021).
- Extended producer responsibility (EPR): the EPR schemes have been introduced by the EU, holding producers financially and operationally accountable for managing the waste of their products, including plastics. This approach fosters the development of environmentally friendly packaging and encourages producers to adopt practices that minimize microplastic generation.
- Waste framework Directive: The water framework directive is a comprehensive legislation addressing water bodies' ecological and chemical status, providing a framework for monitoring, and managing water quality. It includes measures to assess and mitigate microplastic impact in freshwater ecosystems.
- Marine strategy framework directive: The Marine Strategy Framework Directive concentrates on safeguarding the marine environment, incorporating actions to combat

Table 21.2: EU directives towards single-use plastic and microplastic reduction.

EU directives	Date	Objective	EU country participation
Plastic packaging	Entry into force on 31 December 1994. 30 September 2020 to 6 January 2021 was an open public consultation on the review of requirements for packaging and other measures to prevent packaging waste	EU rules on packaging and packaging waste cover all materials, including plastics	Netherlands
Single-use plastics (SUP)	2 July 2019, Commission adopted the Implementing Decision 2021/1752 by 1 October 2021	EU rules on SUPs to fight against marine litter and plastic pollution	Bulgaria
Bio-based, biodegradable, and compostable plastics	First announced 11 December 2019, published action plan on 11 March 2020	The EU will address the sourcing, labeling, and use of bio-based plastics, and the use of biodegradable and compostable plastics	Denmark
Plastic bags	Amendment to the Packaging and Packaging Waste Directive (94/62/EC)	EU rules on plastic bags to address the unsustainable consumption and use of lightweight plastic carrier bags	Ireland, Denmark, France
Global action on plastics	Adopted December 2020, UNEA5.2 will take place from 28 February to 4 March 2022	The EU is paving the way for a global agreement on plastics, to support the global shift towards a circular economy	Germany
Microplastics	The first and second stakeholder meetings held September and November 2021, respectively	The EU aims to address the growing volume of microplastics in the environment	Netherlands, Austria, Belgium, Sweden, Luxembourg
Plastic waste shipments	Entry into force on 1 January 2021	EU rules on importing and exporting plastic waste	Poland

Source: *Adapted from Onyena et al. (2021), Schnurr et al. (2018), Xanthos and Walker (2017).*

marine litter, including microplastics. Member states are mandated to establish measures to achieve or uphold a good environmental status in marine waters, thereby addressing plastic pollution (European Maritime Safety Agency, n.d.).

- Restriction on intentionally added microplastics: The EU has contemplated restrictions on intentionally added microplastics in cosmetics, detergents, and agricultural products to prevent their release into the environment during product use.
- Circular Economy Action Plan: The EU's circular economy action plan underscores the EU's commitment to reducing plastic waste and fostering recycling. Through transitioning to a circular economy model, the EU seeks to minimize plastics' environmental footprint, including the generation of microplastics (European Parliament's Committee on the Environment, n.d.).

Table 21.3 Presents some of the legislative and regulatory policies given by the EU to control microplastic pollution.

Table 21.3: Nonexhaustive examples of global plastic bag bans.

Country	Action plan	Year	Policy	Aim	Enforcement	Tax, levy, fines	Impacts
Africa							
Nigeria	Ban	2019	Plastic Bags Prohibition Bill (not approved by Senate)	Plastic pollution mitigation	Not enforced	Fines of ₦500,000 or imprisonment up to 3 years	Declining interest; potential for significant eco-friendly impact
Benin	Partial ban	2018	Ban on nonbiodegradable plastics	Environmental Protection and sanitation	Poorly enforced	Fine range: 5000–100,000 CFA francs	In effect; Impact assessment too early to determine
Kenya	Ban	2017	National ban on plastic bags (enforced)	Environmental preservation, solid waste management	Enforced	4-year prison or 40,000 KES fine	Reduction rate of about 100 million plastic bags yearly
Ghana	Tentative ban	2014	Attempted ban (not implemented)	Revenue generation for plastic waste management	Poorly enforced	No law in place	Focus on managing plastics rather than banning
Togo	Ban	2011	Ban on nonbiodegradable plastics	Environmental Protection	–	Fines from 5 million to 10 million FCAF or imprisonment	–
Morocco	Ban	2009/ 2016	Ban on manufacture, distribution, and importation of plastic bags	Plastic bag pollution halt	Poorly enforced	Fines from $20,000 to over $100,000	Promoting accessible plastic bag alternatives
Rwanda	Ban	2008	National ban on non-biodegradable plastic bags (poorly enforced)	Plastic pollution mitigation	Poorly enforced	Fines, imprisonment, public confessions, license suspension	Increase in tourism, creating 177,000 new jobs due to reduced plastic bags pollution
South Africa	Tax/ partial ban	2003	Prohibition of Plastic carrier bags, tax on thicker bags (enforced)	Revenue generation, removal of harmful plastic products	Enforced	Levy increased from 12 to 25 cents (April 2020)	Revenue from bag levies increased by R59 million to R300 million between 2018 and 2019

Asia							
Malaysia	Ban	2017	Ban on nonbiodegradable plastic bags (poorly enforced)	Environmental protection and sustainability	Poorly enforced	Customer charges for biodegradable bags, fines for noncompliance	Growing practice
Israel	Ban	2017	Banned distribution of lightweight plastic bags (<20 μm)	Cut plastic waste	–	Charges for bags (20–50 μm) in supermarkets	80% drop in plastic bag consumption
India (Karnataka)	Ban	2016	Ban on single-use plastic items (nonenforced)	Decrease plastic pollution	Nonenforced	Fines in Kolkata (Rs 500), New Delhi (Rs 5000)	Impact studies pending due to weak enforcement
China	Ban	2008	Ban on single-use plastic bags in grocery stores (enforced)	Decrease in plastic bag production	Enforced	Fines of 10,000 yuan for illegal distribution	66% decrease in plastic bag usage; 600,000 regulators sent for compliance
Bangladesh	Ban	2002	Ban on polyethylene bags <55 μm thickness (nonenforced)	Environmental protection and plastic pollution mitigation	Nonenforced	Jail or fine of TK50,000–10 lakh	Lack of impacts due to nonenactment of the law and lack of eco-friendly options
Europe							
France	Ban/tax	2017	Ban on free plastic bags in supermarkets, tax in effect	Plastic waste reduction	–	Tax of €0.04 per bag (rose to €0.07 in 2019)	Too early to assess impacts
Netherlands	Tax	2016	Tax on plastic bags with exemptions for food-related use	Combat litter in streets, prevent wastage of resources	Nonenforced	25 Euro cents per bag advised but not enforceable	40% reduction in plastic bag use
United Kingdom	Tax	2015	5p charge for single-use plastic carrier bags (enforced)	Economic and environmental benefits	Enforced	Levy on consumers (£0.05), voluntary for smaller retailers	Over 80% decline in plastic bag consumption; significant economic benefits for the UK Government
Bulgaria	Tax	2012	Tax for single-use plastic bags (<25 μm)	Waste prevention and management	–	Tax started at 15 stotinki per bag (increased to 55 stotinki)	Bag consumption more than halved in the first month

(*Continued*)

Table 21.3: (Continued)

Country	Action plan	Year	Policy	Aim	Enforcement	Tax, levy, fines	Impacts
Ireland	Tax	2002	Levy for sale of plastic bags in retail stores (enforced)	Reduce plastic waste and adverse effects	Enforced	Levy increased from 15 Euro cents (2002) to 44 Euro cents (2009)	Discarded plastic bags reduced to 0.13% of litter pollution in 2015
Denmark	Tax	1994	Tax on plastic bags based on weight (non-enforced)	Environmental protection	–	Plastic bags cost consumers between 37 and 65 US cents	Reduction in single-use plastic bags
Germany	Tax	1991	Tax on plastic bags in retail stores (enforced)	Environmentally friendly	Enforced	5 or 10 Euro cents per bag (increased to 20 cents)	Reduction in the use of lightweight plastic bags
North America							
Boston, MA	Tax/ban	2018	Dual approach of taxation and bans on single-use bags	Reduction of plastic waste	–	Tax of 5 cents per bag	Over 350 million plastic bags yearly reduced
Canada	Tax	2016	Tax for all shopping bags (non-enforced)	Reduce plastic bag use	Nonenforced	Walmart Canada began charging customers a 5-cent fee	–
Seattle, WA	Ban	2012	Retail stores banned from releasing single-use bags	Reduction of plastic waste	–	Tax of 5 cent per bag	78% reduction in plastic bag use
United States of America	Tax	2009	5-cent tax on plastic bags (enforced)	Curbing Plastic Pollution, Revenue generation	Enforced	A 5-cent tax on plastic bags	85% reduction in plastic bag consumption; significant impact in D.C.
San Francisco	Policy	2007/ 2012	Policy to use reusable bags, 10-cent fee on single-use bags	Zero waste by 2020 and environmental stewardship	–	No bag fee	72% decrease in plastic bag use since 2010
Oceania							
New Zealand	Ban	2019	Ban on plastic shopping bags <70 microns	Plastic pollution mitigation	–	Defaulters to pay fines up to £51,000	Impact is yet to be seen

Papua New Guinea	Ban	2016	Nation-wide ban on plastic bags	Plastic pollution mitigation	–	Companies face K50,000 fine on plastics	Promoted use of traditional and locally manufactured bilum bags
Australia	Ban	2011	Ban on plastic bags (<35 microns)	Reduce plastic waste and green ecosystem	Enforced	–	Ban eliminated 33% of plastic waste sent to landfills
South America							
Chile	Ban	2017 (2018)	Ban on businesses distributing plastic bags, fines enforced	Plastic waste reduction	Enforced	Fines equal to USD 300	Municipalities restricting plastic bag distribution; some areas banned plastic bags
Colombia	Ban	2017	Ban on single-use plastics and bags <30 cm	Reduce single-use plastics	–	A fee of 1 US cent	Too early to assess impacts
Brazil	Ban	2015	Ban on the distribution of plastic bags in supermarkets	Encourage biodegradable bags	Nonenforced	–	Lack of enforcement hinders positive impacts
Argentina	Ban	2012	Ban on plastic bags in Mendoza and Buenos Aires provinces	Discourage nonbiodegradable bags	–	–	Use of biodegradable bags and boosts recycling incentives

21.6.3.1.2 Research and innovation programs

The EU supports research and innovation projects focused on understanding and addressing microplastic pollution. Funding programs encourage the development of technologies for monitoring, prevention, and clean-up of microplastics in various environments.

21.6.3.2 United States

21.6.3.2.1 Federal and state-level actions

U.S. Environmental Protection Agency (EPA) has come up with various national-level strategies to address plastic pollution, like the National Recycling Strategy, presented in Table 21.4.

21.6.3.2.2 International programs conducted by the United States

- CAFTA-Dr and Panama Environment Cooperation: The Environmental Protection Agency (EPA) and the U.S. State Department's Bureau of Oceans and International Environmental and Scientific Affairs (OES) are actively working to improve solid waste management capability. In line with the terms of the U.S.-Panama Trade Promotion Agreement and the U.S.-Dominican Republic-Central America Free Trade Agreement (CAFTA-Dr), this programme attempts to address land-based sources of ocean plastic pollution within Central American countries and the Dominican Republic.
- Increasing West Africa's ability to manage plastic garbage in an environmentally responsible manner: The International Union for Conservation of Nature (IUCN) and the OES Office of Environmental Quality have formed a grant collaboration. Through capacity-building projects in Senegal, their combined efforts are concentrated on lowering the amount of plastic debris that leaks into the maritime ecosystem.
- One workshop under the U.S.-Singapore Third Country Training Programme 2021 focused on the problem of plastic pollution in ASEAN. This programme, which emphasizes the importance of supporting integrated solid waste management methods, seeks to support locally and regionally sustainable solutions.
- Global Ghost Gear Initiative (GGGI): To address marine waste, including ghost gear, in the Caribbean, the State Department has granted funding totaling over $1.5 million to the Ocean Conservancy and GGGI, its affiliate. In addition, the Ocean Conservancy received a grant of around $1 million to help Vietnam with its efforts to combat marine garbage. In 2020, the US formally became a member of GGGI.
- APEC Marine Debris Management and Innovation (MDMI) Sub-Fund: The State Department arranged for further contributions and replenishments from other APEC members after the United States made an initial $800,000 donation. Applications for MDMI project funds to combat marine plastic pollution are open to all nations.

Table 21.4: United States of America directives towards single-use plastic and microplastic reduction.

U.S.A. directives	Date	Objective	Reference
U.S. National Protection Agency (EPA)			Environmental Protection Agency (n.d.)
National Recycling Strategy	November 2021	Reaffirmed the goal to increase the U.S. recycling rate to 50% by 2030.	
Sustainable Material Management (SMM) program	Annual Report	To provide information on Municipal Solid Waste (MSW) generation, recycling, composting, combustion with energy recovery, and landfilling.	
WasteWise Program	–	EPA along with businesses, governments, and nonprofit organizations promote the use and reuse of materials more productively over their entire life cycles. Partners demonstrate how they reduce waste, practice environmental stewardship, and incorporate sustainable materials management into the business model, including waste-handling processes.	
U.S. Department of Energy (DOE)			US Department of Energy (n.d.)
Plastics Innovation Challenge	2018	The goals of this program are to develop solutions that delivery greater than 50% energy savings, address greater than 90% of plastics, reduce greenhouse gas emissions by more than 50%, and achieve at least 75% carbon utilization.	
U.S. Department of Agriculture (USDA)			US Department of Agriculture (n.d.)
Agriculture Research Service	1953	Scientists developed a way for crop residues from cornstalks, straw, and sugarcane bagasse, combined with sustainable enzymes, to convert crop-derived sugars into compounds used in the sustainable manufacture of nylon plastics.	
BioPreferred Program (Farm Security and Rural Investment Act)	2002	Works to increase the purchase and use of biobased products through federal procurement and a certification and labeling initiative.	
U.S. Department of Interior (DOI)			US Department of the Interior (n.d.)
Bureau of Safety and Environment Enforcement (BSEE)	2011	The BSEE Marine Trash and Debris prevention standards require offshore energy companies to conduct annual training for all Outer Continental Shelf (OCS) employees to adopt best practices to reduce marine debris.	

(*Continued*)

Table 21.4: (Continued)

U.S.A. directives	Date	Objective	Reference
National Park Services (NPS)	January, 2022	Along with partners to extend the benefits of natural cultural resource conservation and outdoor recreation throughout this country and the world.	
S. Fish and Wildlife Service (USFWS)		Manages more than 180 coastal national wildlife refuges and five marine national monuments across the United States.	
US Food and Drug Administration (FDA)			US Food and Drug Administration (n.d.)
Recycled plastics for food packaging		Safe usage of recycled plastics for food contact articles.	
National Aeronautical and Space Agency (NASA)		Funded projects to investigate the capability of using satellite remote sensing to detect microplastics in our oceans using hyperspectral remote sensing, an important capability of NASA's upcoming PACE mission.	National Aeronautics and Space Administration (n.d.)
National Institute for Standards and Technology (NIST)		The organization supports U.S. work on documenting standards, reference materials, and reference data and serves as a diverse stakeholder convenor on manufacturing and technology innovation needs in the U.S. economy and conducts work on the thermal and mechanical properties of polymers at different stages in the recycling process.	National Institute of Standards and Technology (n.d.)
National Oceanic and Atmospheric Agency (NOAA) Marine Debris Program	2006	To address the adverse impacts of marine debris on the U.S. economy, the marine environment, and navigation safety.	National Oceanic and Atmospheric Administration NOAA (n.d.)
National Science Foundation (NSF)		Supports basic research that develops fundamental knowledge and engineering advances pertaining to recycling, polymer chemistry and physics, alternative materials, sustainable and circular processes, the fate and impact of plastic materials lost to the environment, and pollution mitigation, control systems, and remediation.	National Science Foundation (n.d.)

21.6.3.2.3 Collaborative efforts with industries

Recognizing the environmental and health impacts of microplastic pollution, the United States government collaborates with industries to address this issue. This partnership combines public and private expertise and resources, focusing on developing sustainable materials and improving recycling technologies to minimize microplastic generation, the government also

works with the textile industry to reduce microfibre shedding and supports research on innovative detection and removal methods for microplastics. Policy initiatives include stricter plastic waste regulations and cleanup project support, highlighting a strategic alignment of government and industry efforts toward sustainability and environmental protection. Table 21.5 delineates the U.S. agencies collaborating with industries for international development.

21.7 Global collaborations

21.7.1 International cooperation on microplastic research

International cooperation on microplastic research has intensified, highlighting the urgency to combat microplastic pollution globally. Collaborations include various countries, research bodies, and international entities. The UN-sponsored GESAMP conducts marine microplastic studies, while the EU's Horizon 2020 funds projects like "RESPONSE" with €8 million to tackle microplastic contamination. The International Research Coordination Network facilitates global data and methodology sharing. This collaborative approach is vital, as the UN Environment Programme's report shows over 800 marine species are impacted by this pollution. These global efforts pool resources and standardize research methods, enhancing efficiency and relevance in addressing microplastic pollution.

Table 21.5: U.S. Agency for International Development (USAID).

International support by United States	Year	Program impact
Clean Cities, Blue Ocean	In 2023	Safely managed over 556,000 metric tons of waste and recyclables. Prevented over 82,000 metric tons of plastics (the equivalent of nearly 6.5 billion plastic bottles) from leaking into the environment. Recovered and diverted over 19,000 metric tons of plastic and other waste from landfills while supporting local livelihoods and building circular economies. Trained over 14,500 individuals from local government, the informal waste sector, and local organizations to build capacity for solid waste management planning and programs. Mobilized over $12.6 million in additional investments for improved solid waste management systems and infrastructure.
Municipal Waste Recycling Program	2016–21	Worked to reduce land-based sources of ocean plastic pollution in four key countries – Indonesia, Philippines, Sri Lanka, and Vietnam.
Private Sector Partnerships	2019	USAID launched an agreement leveraging more than $100 million in a private-sector investment strategy managed by circulated capital and funded by multinational companies. USAID provides a $35 million, 50% loan-portfolio guarantee through the U.S. International Development Finance Corporation (DFC), which is used to incentivize private capital investment in the recycling value chain in South & Southeast Asia.

21.7.2 Partnerships between governments, NGOs, and industries

Partnerships between governments, NGOs, and industries are pivotal in tackling global environmental and sustainability challenges. These collaborations harness diverse expertise, resources, and networks to drive impactful change. An exemplar of such cooperation is the Grow Africa initiative, formed by the World Economic Forum and the African Union, and bolstered by G8 support. This initiative secured over $3 million in private-sector investment for African agriculture, illustrating the transformative power of cross-sector collaboration.

In another notable example, the Healing Forests Initiative, a collaboration between the World Wildlife Fund and Johnson & Johnson, explores the interconnections between deforestation, human health, and economies. This partnership exemplifies how blending expertise from different sectors can lead to a deeper understanding and innovative solutions for complex problems.

The United Nations Environment Programme (UNEP) actively fosters partnerships across levels and sectors, recognizing that environmental challenges require concerted efforts. These collaborations often involve a wide range of stakeholders, including industry leaders, civil society, and governmental bodies, working together towards common goals.

Such partnerships are crucial for realizing sustainable development goals, as they bring together the strengths of different sectors to address environmental issues more effectively. They demonstrate how coordinated efforts can lead to more innovative, sustainable practices, benefiting both the environment and society at large.

21.8 Learnings from the strategies of developed countries for controlling microplastic contamination

Developing countries can glean valuable insights from the strategies implemented by developed nations to control microplastic pollution. First, stringent regulatory frameworks have proven effective. Developed countries have often imposed regulations on the production and disposal of plastic materials, significantly reducing the amount of microplastics entering the environment. Secondly, investments in research and development are crucial. Developed countries have funded extensive studies into the sources, impacts, and mitigation strategies for microplastics, leading to innovative solutions such as improved waste management systems and biodegradable materials. Additionally, public awareness campaigns in developed nations have played a significant role in changing consumer behavior and reducing single-use plastics. Developing countries can also benefit from international collaboration, as sharing expertise and resources can lead to more effective and cost-efficient strategies. Finally, incorporating

microplastics management into broader environmental and sustainable development policies can ensure a more holistic approach to addressing the issue. By adopting and adapting these strategies to their unique contexts, developing countries can make significant strides in combating microplastic pollution.

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